Module 4

1. Student Performance Analysis

A college maintains student records containing student name, Mathematics marks, Science marks, Computer Science marks and attendance percentage.

Write a Python program using Pandas and Matplotlib to perform the following operations:

1. Create a Pandas DataFrame containing student details.
2. Calculate the average marks of each student.
3. Identify the student who scored the highest average marks.
4. Display subject-wise average marks using a bar chart.
5. Represent the percentage of students in different grade categories using a pie chart.
6. Display the marks of a student in different subjects using a line chart.
7. Visualize the distribution of students' average marks using a histogram.
8. Create a scatter plot to represent the relationship between attendance percentage and average marks.

2.Employee Data Analysis

An organization maintains employee details containing employee ID, age, experience, salary and department.

Write a Python program using Pandas and Matplotlib to perform the following operations:

1. Create a DataFrame containing employee details.
2. Calculate the average salary of employees.
3. Identify the employee receiving the highest salary.
4. Display department-wise average salary using a bar chart.
5. Represent the percentage of employees in each department using a pie chart.
6. Display the salary growth of an employee over five years using a line chart.
7. Visualize salary distribution using a histogram.

8. Create a scatter plot to represent the relationship between experience and salary.
9. Use different colours to represent different departments.

3. Weather Data Correlation Heatmap

A weather station records temperature, humidity, rainfall, wind speed and atmospheric pressure.

Write a Python program to:

1. Create a DataFrame containing weather information.
2. Calculate the correlation matrix.
3. Display the correlation heatmap.
4. Identify the two variables having the strongest positive correlation.
5. Identify the variables having negative correlation.

4.Sales Data Correlation Analysis

A supermarket maintains product details containing product price, quantity sold, discount percentage, advertising expenditure and total sales.

Write a Python program to:

1. Create a DataFrame containing sales details.
2. Calculate the correlation between numerical variables.
3. Display the correlation matrix using a heatmap.
4. Identify the variables most strongly correlated with total sales.
5. Suggest which variables are useful for predicting sales.

5. Iris Dataset Correlation Heatmap

Load the Iris dataset from sklearn.datasets.

Write a Python program to:

1. Load the Iris dataset.

2. Convert it into a Pandas DataFrame.
3. Calculate the correlation matrix.
4. Display an annotated heatmap using sns.heatmap().
5. Identify the two most highly correlated features.
6. Plot the two features using a scatter plot.
7. Use different colours to represent the three flower classes.